



Promo codes — Product specification

Product	Cartwheel storefront
Feature	Promo codes
Version	1.0
Status	Approved for development
Last updated	27 Sep 2026

1. Overview

Shoppers can apply a promo code to their cart to get a discount or free shipping, and can check any code on an Offers page without having a cart. The server is the source of truth for every calculation; the web app displays what the server returns.

2. Goals

1. Let shoppers apply one promo code per order on the cart page and see the saving before checkout.
2. Let shoppers check whether a code is valid, what it gives and its conditions, without a cart.
3. Record the applied code and discount on the order and show it on the confirmation page and in Admin → Orders.

3. User stories

ID	As a...	I want to...	So that...
US-1	shopper	enter a code on my cart and see the discount	I know what I'll pay
US-2	shopper	be told exactly why a code doesn't work	I can fix it (e.g. add more items)
US-3	shopper	check a code on the Offers page	I know if it's worth using
US-4	shopper	keep my code when I refresh or come back later	I don't have to enter it again
US-5	store admin	see which code was used on each order	I can track promotions

4. Promo types

Type	Effect
percentage	A percentage of the subtotal, optionally capped at a maximum discount

Type	Effect
flat	A fixed rupee amount off the subtotal
free_shipping	Shipping becomes ₹0
category	A percentage off only the items in one category

5. Launch codes

All dates are in India Standard Time (IST, UTC+05:30).

Code	Type	Value	Min. order	Max. discount	Valid	Limits	Shown on Offers
WELCOME10	percentage	10%	₹999	₹300	Always	Once per email	Yes
FLAT250	flat	₹250	₹1,999	—	Always	—	Yes
FREESHIP	free_shipping	Shipping ₹0	—	—	Always	—	Yes
BOOKS20	category (Books)	20%	—	—	Always	—	Yes
DIWALI25	percentage	25%	₹2,999	₹1,000	15 Oct 2026 – 5 Nov 2026	—	Yes
MONSOON15	percentage	15%	₹499	₹200	1 Jul 2026 – 15 Sep 2026	—	Yes
LIMITED5	flat	₹100	—	—	Always	5 uses in total	No

6. Business rules

Definitions. *Subtotal* is the sum of item price × quantity, before any discount and before shipping. *Shipping* follows SH-01: ₹99 when the subtotal is below ₹999, free when the subtotal is ₹999 or more. *Eligible subtotal* is the subtotal of the items a code applies to (all items, except for `category` codes).

Rule	Statement
BR-01 Normalisation	Leading and trailing spaces are removed. Codes are case-insensitive (<code>welcome10</code> = <code>WELCOME10</code>). Codes are always displayed in upper case.
BR-02 Format	After normalisation a code is 3–20 characters, letters A–Z and digits 0–9 only. Anything else fails with <code>PROMO_INVALID_FORMAT</code> .
BR-03 Input behaviour	The Apply button is disabled while the input is empty or only spaces. Pressing Enter in the input applies the code. While the code is being checked the button shows "Applying..." and is disabled.

Rule	Statement
BR-04 One code per order	Only one code can be applied. Applying a different code replaces the current one (if the new code fails, the current one stays applied).
BR-05 Minimum order	A code with a minimum order applies when the subtotal is greater than or equal to the minimum. Example: a ₹999 subtotal qualifies for WELCOME10. Otherwise it fails with <code>PROMO_MIN_ORDER</code> , and the message states the shortfall.
BR-06 Percentage	Discount = subtotal × percentage, then capped at the code's maximum discount if it has one. Example: WELCOME10 on ₹7,999 → 10% = ₹799.90 → capped to ₹300.
BR-07 Flat	Discount = the flat value, but never more than the subtotal.
BR-08 Free shipping	Shipping becomes ₹0 and the discount is ₹0. If the order already ships free, the code still applies and the message says so.
BR-09 Category	The percentage applies only to the eligible subtotal (items in the code's category). If the cart has no eligible items, the code fails with <code>PROMO_NOT_APPLICABLE</code> .
BR-10 Rounding and totals	The discount is rounded down to a whole rupee and never exceeds the eligible subtotal. Total = subtotal – discount + shipping. Shipping (SH-01) is always based on the subtotal before the discount.
BR-11 Validity window	A dated code is valid from 00:00:00 IST on its start date through 23:59:59 IST on its end date, inclusive . Before the start: <code>PROMO_NOT_STARTED</code> . After the end: <code>PROMO_EXPIRED</code> .
BR-12 Usage limits	A total-use limit fails with <code>PROMO_LIMIT_REACHED</code> once reached. A once-per-email code fails with <code>PROMO_ALREADY_USED</code> if an order with that email already used it; this is checked when the order is placed (the email is not known on the cart). A use is counted only when an order is placed.
BR-13 Order of checks	Checks run in this order and the first failure is reported: format → exists → started → not expired → usage limit → once per email (only when an email is supplied) → cart not empty → minimum order → category eligibility.
BR-14 Cart changes	Whenever the cart changes, the applied code is re-checked and the discount recalculated. If the code no longer qualifies it is removed and the shopper sees "{CODE} was removed. {reason}", where {reason} is the failure message.
BR-15 Persistence and server check	The applied code is saved with the cart and survives a page refresh. The server checks the code again when the order is placed; if it fails, the order is not placed, the code is removed, and the reason is shown on the checkout page.
BR-16 Offers page	<code>/offers</code> lists every code marked "Shown on Offers" as a card: code, benefit, conditions, status badge (Active / Not started / Expired), "Ends in N days" when an active dated code ends within 7 days ("Ends today" on the last day), and a Copy button. A "Check a code" box accepts any code, including ones not listed, and shows its type, benefit, conditions and status — no cart needed.

7. Messages

{CODE} is the normalised code. Amounts use the store format (₹1,299 , or ₹1,299.50 when there are paise). Dates use d MMM yyyy (e.g. 15 Oct 2026).

Error code	HTTP	Message
PROMO_INVALID_FORMAT	400	Promo codes contain only letters and numbers.
PROMO_NOT_FOUND	404	We couldn't find the code {CODE}.
PROMO_NOT_STARTED	422	{CODE} starts on {start date}.
PROMO_EXPIRED	422	{CODE} expired on {end date}.
PROMO_LIMIT_REACHED	422	{CODE} has reached its usage limit.
PROMO_ALREADY_USED	422	{CODE} has already been used with this email.
PROMO_EMPTY_CART	422	Add items to your cart to use a promo code.
PROMO_MIN_ORDER	422	Add ₹{shortfall} more to use {CODE}.
PROMO_NOT_APPLICABLE	422	{CODE} isn't valid for items in your cart.

Situation	Message
Applied (discount)	{CODE} applied. You saved ₹{saving}.
Applied (free shipping, shipping was ₹99)	{CODE} applied. You saved ₹99.
Applied (free shipping, already free)	{CODE} applied. Your order already ships free.
Removed after a cart change or at checkout	{CODE} was removed. {reason}
Network or unexpected error	Couldn't apply code. Please try again.

8. Screens

8.1 Cart (/cart)

- The order summary has a "Have a promo code?" button that reveals the promo box (it starts open when a code is applied).
- Promo box: text input labelled "Promo code", an **Apply** button, and a message line below them.
- When a code is applied, "{CODE} applied" and a **Remove** link appear above the input. The input stays available so another code can replace it (BR-04). Removing restores the prices without a code.
- The summary rows are: Subtotal · Promo · Shipping · Total. The Promo row appears only when the discount is more than ₹0. Its label is "Promo ({CODE} · {short benefit})" and its value is the discount with a minus sign, e.g. "Promo (WELCOME10 · 10% off) –₹300". Short benefits: 10% off , ₹250 off , 20% off Books , Free shipping .
- On phones the whole summary, including the promo message and the Promo row, must be readable without horizontal scrolling and must not be covered by the sticky checkout bar.

8.2 Checkout (/checkout)

- The summary repeats the Promo row (read-only, no input).
- Place Order sends the code; see BR-15 for failures.

8.3 Order confirmation (/order/:id) and Admin → Orders

- Confirmation shows the Promo row when the order has a discount.
- Admin → Orders has **Promo** and **Discount** columns ("—" when none).

8.4 Offers (/offers)

- Linked from the main navigation as "Offers".
- "Check a code" box at the top; offer cards below (one column on phones, two from 768 px, three from 1024 px).
- The code on each card must be fully readable at every screen width.
- Copy puts the code on the clipboard and the button says "Copied" for two seconds.

9. API

All amounts are integer **paise** (₹1 = 100).

GET /api/promos

200 { promos: PromoView[] } — only codes shown on Offers.

GET /api/promos/:code

200 PromoView for any existing code, 400 for bad format, 404 for unknown.

```
PromoView {
  code, type, value,           // value: percent for percentage/category, paise for
flat, 0 for free_shipping
  minOrder, maxDiscount,      // paise or null
  category,                   // "Books" or null
  startsOn, endsOn,           // "YYYY-MM-DD" (IST) or null
  status,                     // "active" | "not_started" | "expired" |
"limit_reached"
  daysLeft,                   // whole days until endsOn (0 on the last day) or null
  benefit, shortBenefit,      // e.g. "10% off, up to ₹300", "10% off"
  conditions                   // e.g. ["Min. order ₹999", "Once per customer"]
}
```

POST /api/promos/apply

Body { code, items: [{ productId, quantity }], email? }. 200 { code, type, shortBenefit, subtotal, discount, shipping, total, savings, message } where savings = discount + (shipping before code – shipping after code). Errors: { errorCode, message } plus shortfall (paise) for PROMO_MIN_ORDER. Item errors (unknown product, bad quantity, not enough stock) use the order API's error format.

POST /api/orders

Accepts an optional promoCode. The order stores promoCode (or null) and discount (paise). If the code fails, responds with the promo error ({ errorCode, message }) and places no order.

10. Acceptance criteria

ID	Given	When	Then
AC-01	Cart with Pulse Wireless Earbuds and The Pragmatic Checklist (subtotal ₹1,498)	apply <code>welcome10</code>	discount ₹149 (₹149.80 rounded down), total ₹1,349, message "WELCOME10 applied. You saved ₹149."
AC-02	Cart subtotal ₹999 (one Pulse Wireless Earbuds)	apply WELCOME10	discount ₹99, total ₹900
AC-03	Cart with two Steel Water Bottles (subtotal ₹898)	apply WELCOME10	error "Add ₹101 more to use WELCOME10."
AC-04	Cart with Hush ANC Headphones (₹7,999)	apply WELCOME10	discount ₹300
AC-05	Cart with one Stoneware Mug Set (₹649)	apply FREESHIP	shipping ₹0, total ₹649, message "FREESHIP applied. You saved ₹99."
AC-06	Cart with a book (₹599) and a mug set (₹649)	apply BOOKS20	discount ₹119 (20% of ₹599 = ₹119.80, rounded down)
AC-07	Cart with only electronics	apply BOOKS20	error "BOOKS20 isn't valid for items in your cart."
AC-08	Today is before 15 Oct 2026	apply DIWALI25	error "DIWALI25 starts on 15 Oct 2026."
AC-09	Server time 5 Nov 2026 23:59:59 IST, cart with Stride Smartwatch (₹4,499)	apply DIWALI25	applies: discount ₹1,000 (25% = ₹1,124.75, capped)
AC-10	Server time 6 Nov 2026 00:00:00 IST	apply DIWALI25	error "DIWALI25 expired on 5 Nov 2026."
AC-11	Cart with one Steel Water Bottle (₹449, below MONSOON15's ₹499 minimum)	apply MONSOON15	error "MONSOON15 expired on 15 Sep 2026.", not the minimum-order message (BR-13)
AC-12	Cart with Arc Desk Lamp (₹1,849) and Steel Water Bottle (₹449), FLAT250 applied	remove the bottle	"FLAT250 was removed. Add ₹150 more to use FLAT250."
AC-13	WELCOME10 applied, page refreshed	—	WELCOME10 still applied
AC-14	An order with email a@example.com used WELCOME10	place another order with WELCOME10 and the same email	order not placed; "WELCOME10 was removed. WELCOME10 has already been used with this email."
AC-15	LIMITED5 used by 5 orders	apply LIMITED5	error "LIMITED5 has reached its usage limit."
AC-16	—	check <code>limited5</code> on /offers	shows ₹100 off, Active

ID	Given	When	Then
AC-17	—	apply WEL-10	error "Promo codes contain only letters and numbers."

11. Out of scope

Stacking multiple codes · codes tied to user accounts · admin screens to create or edit codes · gift cards · referral codes.